

Question Number	Answer	Mark
6(a)	The only correct answer is C Y <i>A is not correct because W does not contain mitochondria</i> <i>B is not correct because X does not contain mitochondria</i> <i>D is not correct because Z does not contain mitochondria</i>	(1)

Question Number	Answer	Additional guidance	Mark
6(b)(i)	A description that makes reference to the following points: - named organelle involved in protein production (1) - role of organelle described in the production of the enzyme (1)	mitochondria / nucleus / ribosomes / {rough endoplasmic reticulum / rER} / Golgi (apparatus/body) e.g. mitochondria – produce ATP for protein synthesis (of enzyme) nucleus – site of transcription of enzyme gene / location of gene for enzyme ribosomes -site of protein synthesis / where polypeptide is formed rER -site of protein synthesis / where polypeptide is formed / formation of 3° structure / packaging protein into (transport) vesicle Golgi apparatus – modification of {protein / enzyme} / packaging {protein / enzyme} into (secretory) vesicle	(2)

Question Number	Answer	Additional guidance	Mark
6(b)(ii)	800		(1)

Question Number	Answer	Additional guidance	Mark
6(b)(iii)	An explanation that makes reference to four of the following points: - doubling the acrosin activity of the sperm cell increases the percentage of egg cells fertilised by 68% / non-linear increase (because) higher acrosin activity means the sperm cells can digest through all of the {outer layer / zona pellucida} (of more egg cells) (1) (allowing) sperm (cells) to {bind to (egg cell) membrane / enter egg (cell)} (1) (so that) sperm nucleus can fuse with egg (cell) nucleus / fusion (of nuclei) can occur (1) (low acrosin activity) could result in death of sperm cells before fertilisation could occur (1)	ACCEPT greater increase (in percentage of egg cells fertilised) between 2.5 and 3(a.u.) ACCEPT 100% fertilisation at 5(a.u.) compared to 32% at 2.5(a.u.) ACCEPT digesting the outer layer {faster /more efficiently} / more digestion occurs Accept converse ACCEPT sperm can reach egg (cell) nucleus ACCEPT sperm can fuse with egg (cell) ACCEPT converse	(4)

Question Number	Answer	Additional guidance	Mark
6(c)(i)	image size divided by magnification	ACCEPT $4.5 \pm 1 \div 200$ ignore unmanipulated equation	(1)

Question Number	Answer	Additional guidance	Mark
6(c)(ii)	An answer that makes reference to three of the following points: • (when a sperm entered egg cell) cortical granules have fused to cell surface membrane / cortical {reaction/enzymes} resulted in hardening of zona pellucida (1) • as zona pellucida (of some egg cells) are damaged there are areas where it is not {present / hardened} (1) • resulting in polyspermy / {an extra / two} sperm have entered (the egg cell) (1)	ACCEPT some zona pellucida is not hardened reject 3 sperm have entered egg cell	(3)